Medical Image Analysis with Federated Learning

BlooMNIST: Variant 8: Federated Learning using Flower

Course: AI for Connected Industries (AI4CI)
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1 Introduction

Federated Learning (FL) enables global model training across distributed data sources without sharing raw data. This is a critical requirement in healthcare, where patient records are subject to strict privacy regulations such as HIPAA and GDPR. This report presents experiments on blood cell classification using the BloodMNIST dataset, exploring how data heterogeneity, model architecture, image resolution, aggregation strategy, and data poisoning attacks affect FL performance.

2 Dataset and Experimental Setup

2.1 BloodMNIST Dataset

BloodMNIST is part of the MedMNIST v2 benchmark [1]. It contains RGB (3 channels) microscopy images of eight individual normal peripheral blood cell types: basophil, eosinophil, erythroblast, immature granulocyte, lymphocyte, monocyte, neutrophil, and platelet. The official split comprises 11,959 training, 1,712 validation, and 3,421 test samples at resolutions of 28×28 and 64×64 pixels.

BloodMNIST is a natural fit for federated learning in a healthcare context. Blood cell classification is a task performed across independent clinical laboratories, each with their own isolated images and confidential patient data.

2.2 Deployment on GitHub

For reproducibility and access, the base files for implementing the whole lab are hosted on GitHub. The readme provides info for setting up the configuration and running the models as deviced. (Git Repo: dfl-mednist-tp)

2.3 Federated Learning Setup

All experiments use the Flower simulation framework with the following default configuration:

Table 1: Default hyperparameter configuration.

Parameter	**Value**
Number of clients	5
Communication rounds	20
Local epochs per round	1
Learning rate	0.01 (SGD, momentum 0.9)
Batch size	64
Image resolution	28×28
Random seed	42
Model	CNN

Five clients were chosen as the smallest number that produces meaningful aggregation behavior while being computationally cost effective (especially for server use on Google Colab). Twenty rounds were sufficient to observe convergence or plateau in all experiments. One local epoch per round minimizes client drift and follows the standard FedAvg standards. [2]. Learning rate of 0.01 and momentum 0.9 are also standard clinical settings for image classification.

2.4 Data Partitioning

The Dirichlet partitioning strategy was implemented to simulate IID and non-IID conditions. It is applied class-by-class, for each of the 8 blood cell classes, a proportion vector of length 5 is sampled, and the class indices are split accordingly. Because the split is per-class, the method handles multi-class datasets naturally without treating the label space as a single entity, which would otherwise conflate different class imbalance patterns.

Alpha value controls the extent of the Dirichlet distribution. (refer to Figure 1)

- $\alpha = 0.01$: high heterogeneity where each client receives samples almost exclusively from one or two classes. Represents the worst-case clinical scenario where one hospital only sees rare subtypes.
- $\alpha = 1$: moderate heterogeneity — the standard benchmark value used in the original FedProx and FedAvg non-IID papers.
- $\alpha = 100$: effectively IID, the proportions are nearly uniform, used as the clean upper-bound reference.

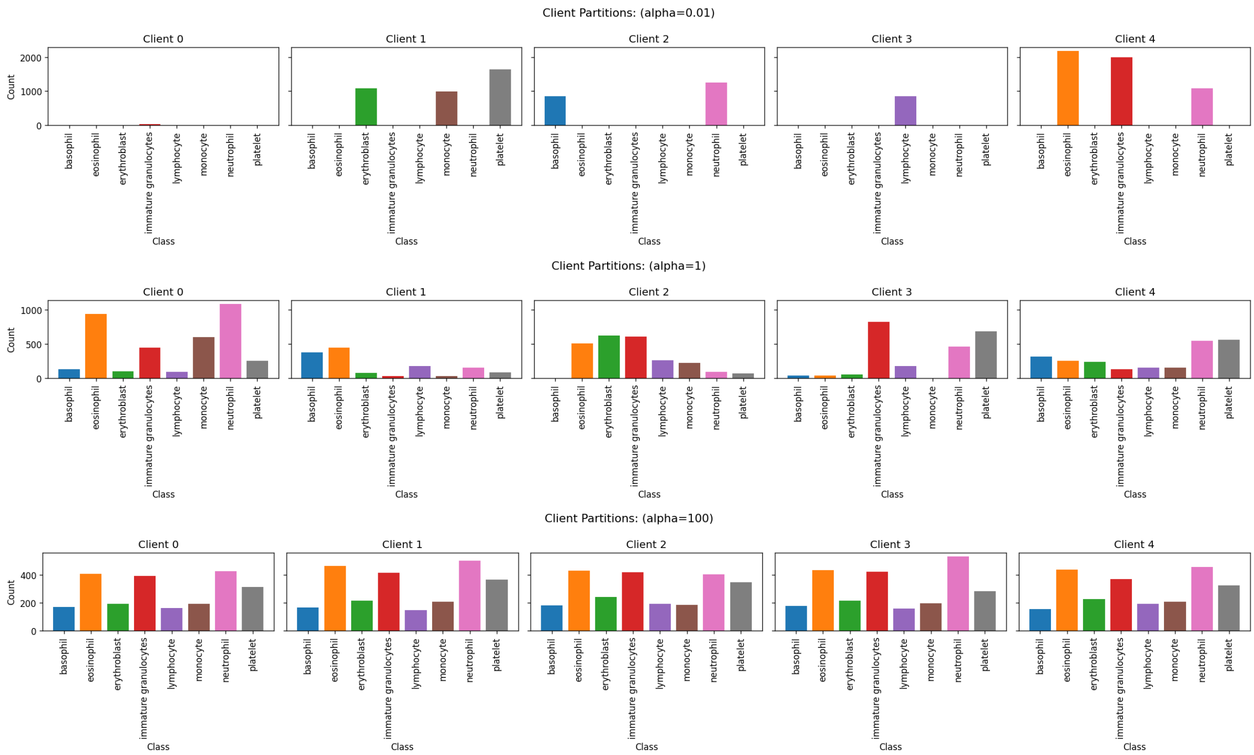

Figure 1: Dirichlet Partitioning for 3 values of Alpha.

2.5 Model Architectures

CNN: Three convolutional blocks (Conv-BN-ReLU), each followed by pooling, with an adaptive average pool to 4×4 and two fully connected layers (≈200K parameters). This represents an edge medical device with limited compute.

ResNet18: Standard torchvision ResNet18 (≈11M parameters) with two architectural adaptations for small inputs: at 28×28 the first convolution is replaced with a 3×3 kernel and the max-pool layer is removed to prevent spatial collapse. The final fully connected layer is replaced to output 8 classes.

The two architectures are practical trade-offs for FL on embedded or resource-constrained clients. The CNN model represents a client that can train quickly with limited compute. ResNet18 represents a more capable hospital server or workstation client where accuracy is the priority. Comparing them under the same FL setup illustrates how model capacity interacts with communication efficiency.

2.6 Aggregation Strategies

Four aggregation strategies were compared:

- **FedAvg**: coordinate-wise weighted mean of client updates. Standard baseline.
- **FedProx** [3]: adds a proximal term $\frac{\mu}{2}\|w - w_{\text{global}}\|^2$ ($\mu = 0.1$) to each client's local objective, penalizing drift from the global model. In the context of BloodMNIST, where different laboratories may hold entirely different cell type distributions, the proximal term provides a theoretically grounded mechanism to counter heterogeneity.
- **FedMedian**: coordinate-wise median, naturally downweighting outlier updates without requiring knowledge of attacker count.
- **Krum** [4]: selects the single client update geometrically closest to its neighbors using Euclidean distances, providing a strong Byzantine defense when the number of malicious clients is known.

2.7 Evaluation Metrics

Validation loss, accuracy, and macro F1 are recorded at every round. Additionally, the global model is evaluated on the held-out test set after every round by loading the aggregated parameters into a proxy model on the server. Macro F1 is used alongside accuracy because non-IID partitioning creates severe per-client class imbalance, and accuracy alone would be misleading if rare classes were systematically ignored.

3 Experiments and Results

3.1 Experiment 1: IID vs. Non-IID Heterogeneity

Setup: FedAvg, CNN, 28×28, comparing $\alpha = 100$ (near-IID) against $\alpha = 0.01$ (extreme non-IID) over 20 rounds.

Table 2: Experiment 1 final results: IID vs. Non-IID (FedAvg, CNN, 28×28).

Condition	Val Loss	Val Acc	Val F1	Test Loss	Test Acc	Test F1
IID $\alpha = 100$	0.250	91.2%	0.895	0.242	92.1%	0.908
Non-IID $\alpha = 0.01$	0.774	68.6%	0.357	0.790	68.3%	0.579

Observations: IID training converges smoothly, while Non-IID training ($\alpha = 0.01$) is severely degraded and results in a large gap in F1 metric. (refer to Figure 2) Convergence curves are highly unstable under extreme non-IID, oscillating between rounds without a clear downward trend in loss.

Interpretation: At $\alpha = 0.01$, each client holds samples from essentially one or two target classes. Each local gradient step pushes the global model towards a local optimum interfering with global updates. The large accuracy-F1 gap (0.686 vs 0.579) under non-IID confirms that the model neglects rare classes entirely, underscoring why macro F1 is the more informative metric in heterogeneous FL.

3.2 Experiment 2: CNN vs. ResNet18

Setup: FedAvg, IID ($\alpha = 100$), 28×28, 20 rounds.

Table 3: Experiment 2 final results: model comparison.

Model	Val Loss	Val Acc	Val F1	Test Loss	Test Acc	Test F1
CNN	0.368	86.8%	0.843	0.350	87.1%	0.852
ResNet18	0.227	92.6%	0.911	0.241	92.3%	0.913

Observations: ResNet18 achieves 92.3% test accuracy and F1 of 0.913, outperforming the baseline CNN. Both models show non-monotone convergence typical of FL in IID conditions. As rounds progress, Resnet18 consistently hold higher peaks. (refer to Figure 3)

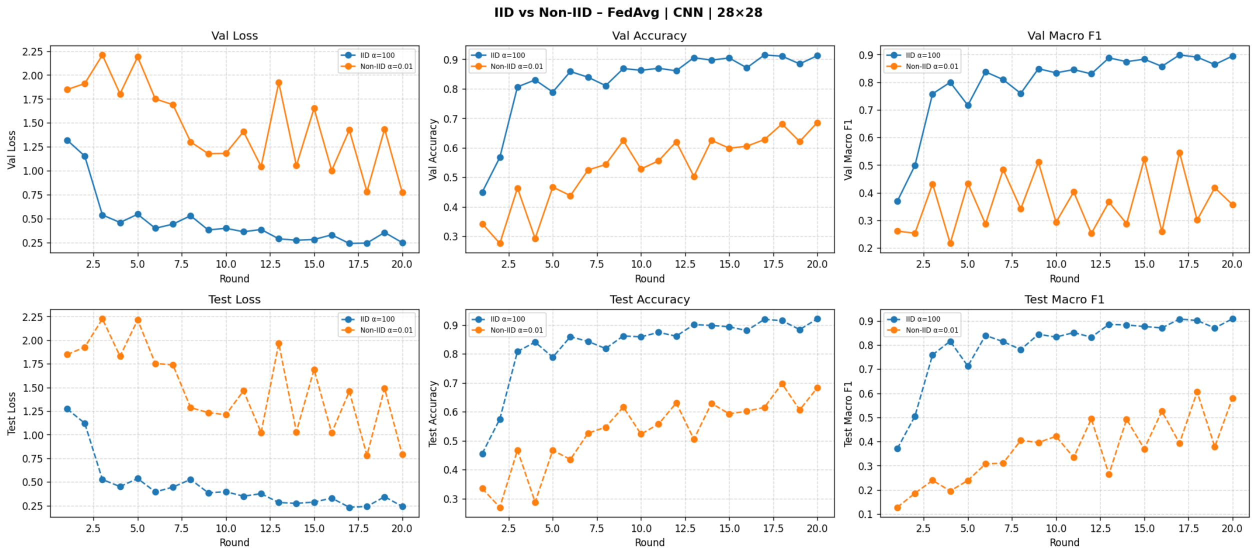

Figure 2: IID vs. Non-IID convergence curves (FedAvg, CNN, 28×28).

Interpretation: The gap is meaningful yet moderate given the FL setting. With only ∼2,400 training samples per client (IID), ResNet18's larger capacity does not overfit, as test accuracy closely tracks validation accuracy. CNN converges slightly faster in the early rounds due to a lower parameter count, which reduces computational overhead. It also offers a practical choice for resource-constrained edge clients while ResNet18 is preferred when accuracy is the priority and clients have sufficient computational budget.

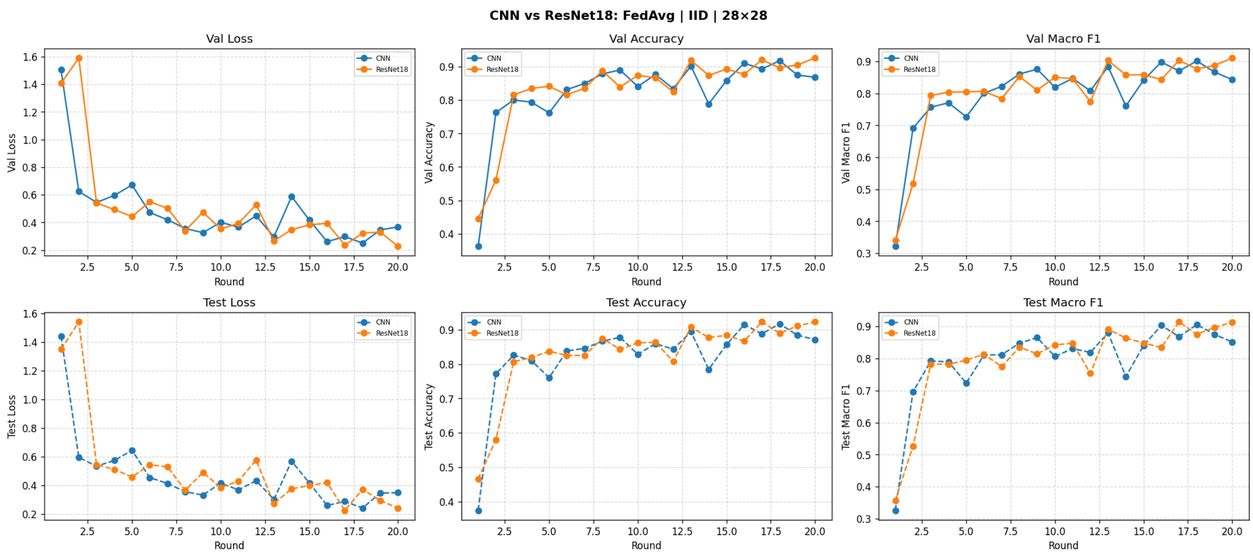

Figure 3: CNN vs. ResNet18 under federated IID training.

3.3 Experiment 3: Aggregation Strategy Comparison

3.3.1 Experiment 3a: Comparison under IID conditions

Setup: CNN, IID ($\alpha = 100$), 28×28, all four strategies, 20 rounds.

Observations (IID): Under clean IID conditions the strategies follow the below performance rank: **FedProx** > **FedMedian** > **FedAvg** > **Krum** (refer to Figure 4)

Krum lags substantially as its selection of a single representative update per round discards useful gradient information from all other clients, significantly distorting convergence. Multi-Krum offers a

Table 4: Experiment 3 final results: strategy comparison (IID, CNN, 28×28).

Strategy	Val Loss	Val Acc	Val F1	Test Loss	Test Acc	Test F1
FedProx	0.258	90.0%	0.879	0.245	90.6%	0.888
FedMedian	0.328	88.0%	0.860	0.338	87.4%	0.859
FedAvg	0.399	85.5%	0.844	0.419	85.4%	0.834
Krum	0.924	75.3%	0.757	0.899	76.1%	0.764

practical alternative.

Interpretation (IID): FedProx's proximal regularization provides a mild benefit even under IID conditions by preventing the local model from drifting far from the global model during the single local epoch. Slightly better FedMedian performance over fedavg could indicate presence of outliers in gradients that mean based model is more influenced by and also vanilla fedavg overfitting to client data. The strategy comparison confirms that the usage of Krum is a trade-off between robustness and efficiency. Hence, should only be preferred when Byzantine robustness is required.

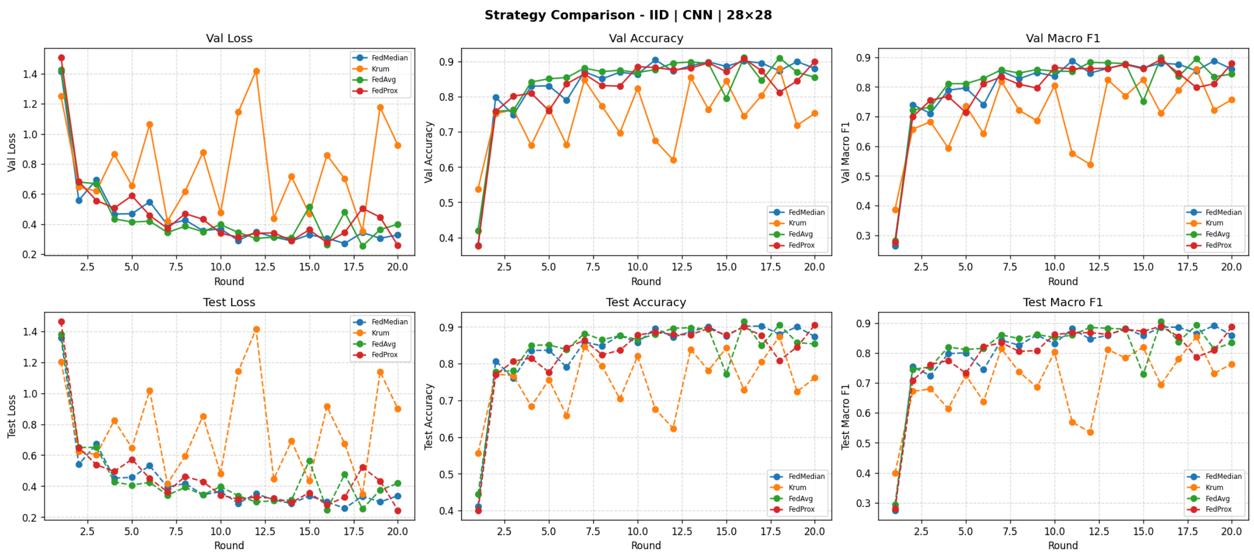

Figure 4: Aggregation strategy comparison under clean IID conditions.

3.3.2 Experiment 3b: Comparison under Non-IID conditions

Setup: CNN, non-IID ($\alpha = 0.01$), 28×28, all four strategies, 20 rounds.

Table 5: Experiment 3b final results: strategy comparison (Non-IID $\alpha = 0.01$, CNN, 28×28).

Strategy	Val Loss	Val Acc	Val F1	Test Loss	Test Acc	Test F1
FedAvg	0.772	71.8%	0.426	0.776	71.8%	0.563
FedProx	0.884	71.2%	0.355	0.897	70.4%	0.505
FedMedian	1.567	40.4%	0.248	1.550	41.1%	0.281
Krum	8.053	27.0%	0.234	8.091	26.5%	0.099

Observations (non-IID): Under non-IID conditions the ranking from Experiment 3 reverses sharply: **FedAvg** > **FedProx** > **FedMedian** > **Krum** (refer to Figure 5)

Interpretation (non-IID): The assumptions of Coordinate-wise median and single-update selection collapses under $\alpha = 0.01$ non-IID, as each client's update points in different directions affecting the global signal. FedAvg and FedProx, by averaging across all clients, retain information from every client's gradient. Although FedAvg has better final metrics, FedProx is has relatively stable learning that has potential to improve with increasing rounds.

The result highlights an important trade-off: FedMedian and Krum's Byzantine robustness (Experiment 5) comes at a severe cost under benign but heterogeneous conditions, where simple averaging is more robust in no attack heterogeneous conditions.

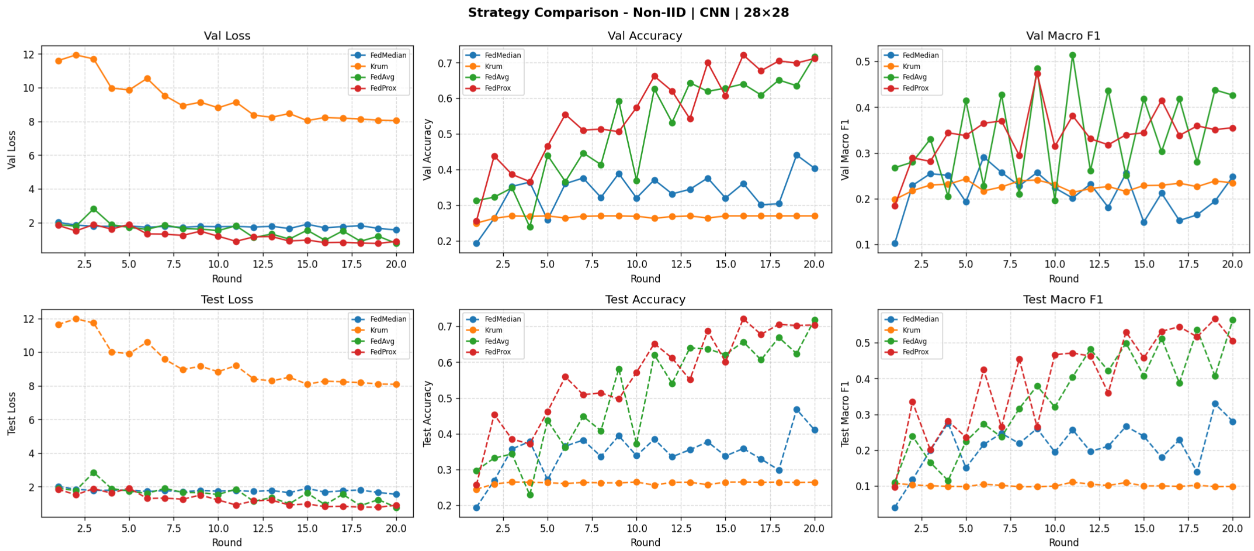

Figure 5: Aggregation strategy comparison under clean non-IID conditions.

3.4 Experiment 4: Image Resolution

Setup: FedAvg, CNN, non-IID ($\alpha = 0.01$), comparing 28×28 vs. 64×64, 20 rounds.

Table 6: Experiment 4 final results: resolution comparison (FedAvg, CNN, non-IID $\alpha = 0.01$).

Resolution	Val Loss	Val Acc	Val F1	Test Loss	Test Acc	Test F1
28×28	1.032	59.0%	0.293	1.027	59.7%	0.515
64×64	1.047	60.4%	0.295	1.050	60.2%	0.408

Observations: The difference between resolutions is marginal as 64×64 achieves only 0.5 percentage points higher test accuracy. Both resolutions show unstable convergence under non-IID conditions, with loss oscillating across rounds and no clear plateau. (refer to Figure 6)

Interpretation: The primary discriminative features across blood cells are adequately captured at 28×28 resolution. The marginal accuracy gain from 64×64 does not justify the high per-sample compute cost and local storage requirements on client devices. The lower F1 at 64×64 despite higher accuracy suggests possible heterogeneous clients overfitting to locally dominant classes more aggressively. For federated edge deployment, 28×28 is the recommended resolution.

3.5 Experiment 5: Data Poisoning Robustness

Setup: CNN, 28×28, non-IID ($\alpha = 0.5$), 20% malicious clients performing random label-flipping. Strategies compared: FedAvg (no defense), FedMedian, Krum, and baseline FedAvg.

Observations: The most significant aspect is the deviation between validation and test metrics. The gap for FedMedian occurs because the malicious client reports high accuracy on its own flipped-label partition while inflating the weighted-average val metric. The held-out test set reveals the true generalization quality. FedMedian fairly recovers from the attack, although krum mostly suffers from heterogeneous conditions with large dips. (refer to Figure 7)

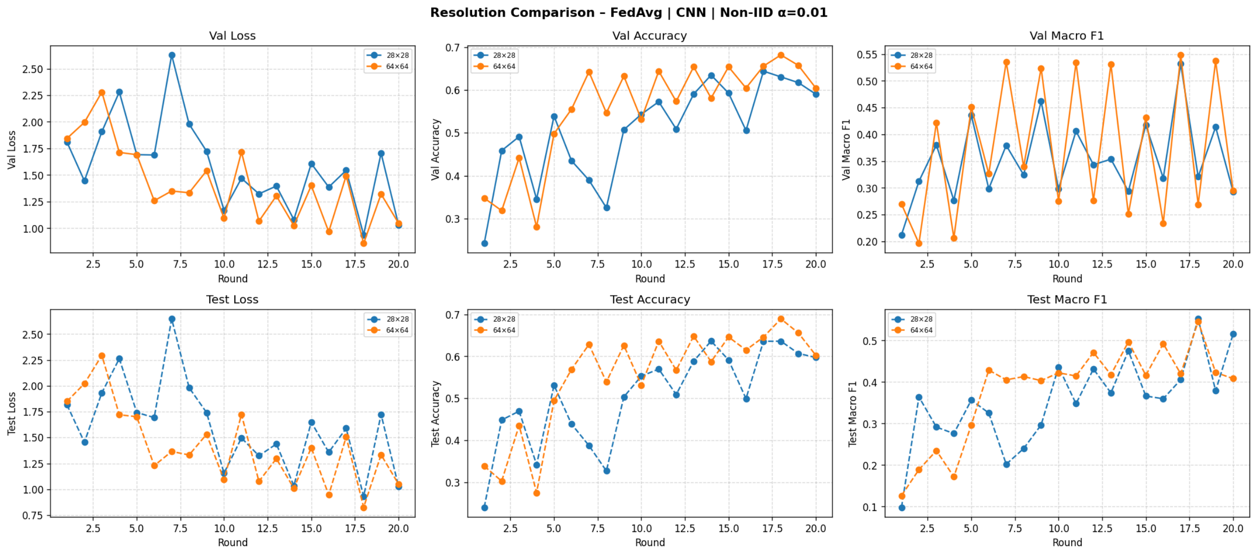

Figure 6: Resolution comparison under non-IID conditions.

Table 7: Experiment 5 final results: data poisoning (non-IID $\alpha = 0.5$, 20% malicious).

Condition	**Val Loss**	**Val Acc**	**Val F1**	**Test Loss**	**Test Acc**	**Test F1**
FedAvg clean	0.372	86.4%	0.700	0.355	87.3%	0.843
FedMedian poisoned	1.046	76.3%	0.573	0.445	85.2%	0.818
FedAvg poisoned	0.915	75.4%	0.592	0.512	84.2%	0.817
Krum poisoned	2.293	65.8%	0.501	1.593	72.6%	0.646

Interpretation: FedMedian's coordinate-wise median naturally suppresses the malicious update, maintaining test performance close to the baseline. FedAvg's test accuracy under poisoning is less affected, because with only 1 attacker out of 5 clients, the poisoned gradient is partially diluted by the four honest updates. Krum underperforms because even though it selects a single non-malicious update, discarding the three other honest updates slows convergence, particularly under non-IID heterogeneity. The val/test divergence observed is a useful diagnostic of presence of malicious client updates in the system.

3.6 Experiment 6: Client Drift due to Local Epochs

Setup: FedAvg, CNN, 28×28, local epochs $\in \{1, 5\}$, both IID ($\alpha = 100$) and non-IID ($\alpha = 0.01$).

Table 8: Experiment 6 final results: local epochs and client drift.

Condition	**Val Loss**	**Val Acc**	**Val F1**	**Test Loss**	**Test Acc**	**Test F1**
IID, $e = 5$	0.190	93.5%	0.925	0.197	93.8%	0.931
IID, $e = 1$	0.303	88.9%	0.867	0.302	89.7%	0.879
Non-IID, $e = 5$	0.800	71.8%	0.274	0.814	73.1%	0.668
Non-IID, $e = 1$	0.964	65.1%	0.357	0.935	65.2%	0.497

Observations: Under IID conditions, increasing local epochs from 1 to 5 improves test accuracy and F1 demonstration the clear benefit of federated learning. Under non-IID conditions, the output reverses in terms of F1, $e = 5$ achieves higher test accuracy but substantially lower val F1. The non-monotone val F1 behavior at $e = 5$ non-IID indicates that the model gains accuracy on dominant classes at the expense of rare ones. (refer to Figure 8)

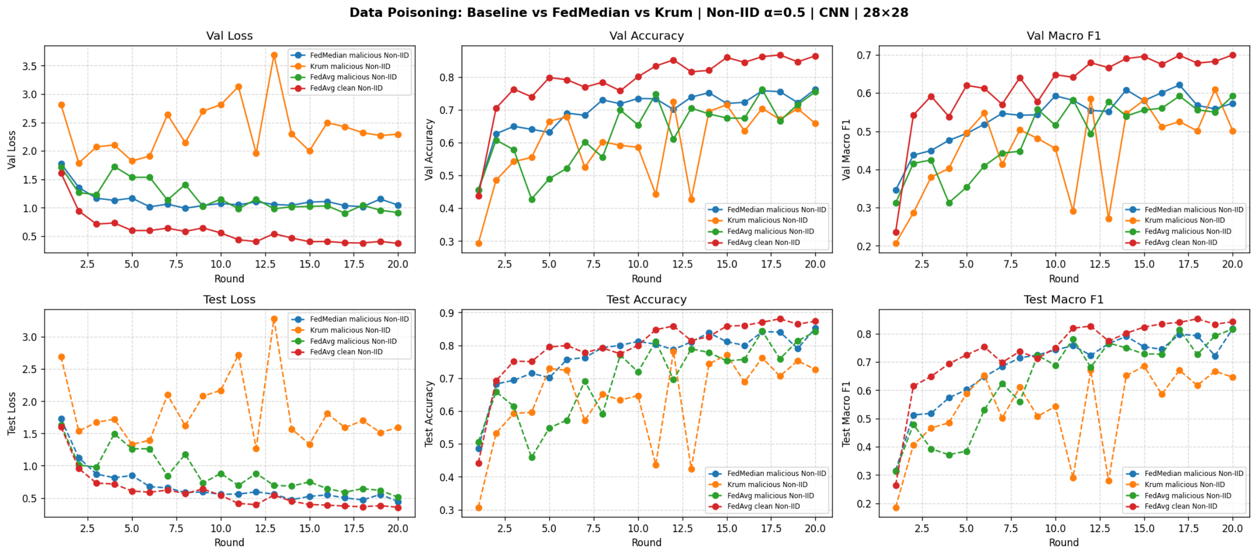

Figure 7: Poisoning robustness: FedMedian and Krum vs. FedAvg baseline.

Interpretation: More local epochs accelerate convergence under IID conditions with slight over corrections in gradients. Under non-IID, the additional local steps cause each client to overfit its local class distribution before aggregation due to client drift. This confirms that local epoch count should be kept low (ideally 1) under heterogeneous data distributions, or that a drift-correcting strategy such as FedProx should be used if more local computation is desired.

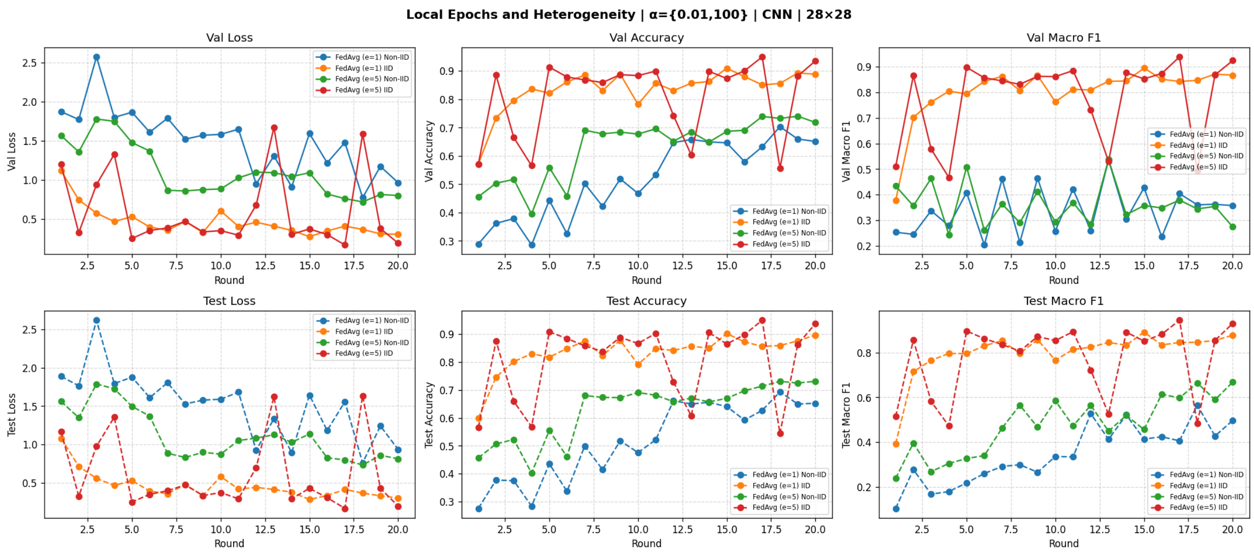

Figure 8: Effect of local epochs on client drift under IID and non-IID conditions.

4 Challenges and Discussion

Accounting multi-class in Dirichlet Partitioning. The approach for non-IID FL simulation draws a vector of proportions from Dirichlet(α) and uses those proportions to assign class samples to clients. This is applied class-by-class for each of the 8 blood cell classes, and the class indices are split accordingly. The result is that each client ends up with a skewed but non-zero representation of most classes when α is not extremely small. Because the split is per-class, the method handles multi-class datasets naturally without treating the label space as a single entity, which would otherwise conflate different class imbalance patterns.

Validation metrics are misleading under poisoning. The weighted-average validation metric aggregated across clients was inflated by the malicious client's high accuracy on its own flipped labels. Adding a server-side test evaluation to score the global model on the clean held-out set each round exposed the true generalization quality independently of client-reported metrics.

Low divergence between Validation and Test metrics across experiments. BloodMNIST is a well-curated, and balanced benchmark dataset. Unlike real-world medical data, it was specifically designed for standardized evaluation, and train/val/test splits come from the same underlying distribution with similar class proportions. As a result, val loss is already a reasonable proxy for generalization.

5 Conclusions

Six federated learning experiments on BloodMNIST yield the following key findings:

1. **Data heterogeneity is the primary challenge.** Extreme non-IID partitioning compared to IID conditions, demonstrates unstable convergence across rounds. Macro F1 was a more sensitive indicator of degradation than accuracy.
2. **ResNet18 outperforms CNN but at a compute cost.** ResNet18 achieved better test accuracy under IID conditions. However, CNN is a practical choice for resource-constrained clients.
3. **FedProx is the best all-round strategy under clean conditions.** Its proximal regularization provided a consistent benefit over FedAvg and FedMedian, while Krum's single selection mechanism is redundant in the absence of attackers. Even in non-IID conditions, FedProx offers stable learning (lower variance across rounds) relative to vanilla FedAvg.
4. 28×28 **resolution is sufficient.** The accuracy gain from 64×64 was under 1%, with lower F1, making 28×28 the preferred choice for federated edge deployment.
5. **FedMedian provides robust defence against label-flipping poisoning.** With 20% malicious clients, FedMedian maintained better accuracy relative to the baseline, while Krum's conservative selection yielded low gains. The val/test accuracy gap is a useful real-time diagnostic for Byzantine attack presence.
6. **Local epoch count must be managed under heterogeneity.** Five local epochs improved IID accuracy but suppressed macro F1 for rare classes under non-IID. One local epoch per round is recommended for heterogeneous deployments.

References

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